Categorical composable cryptography

Anne Broadbent, Martti Karvonen

FoSSaCS 2022

Composability in cryptography

One would expect that if you wire together provably secure protocols you end up with a secure protocol.

- This is false in general! Standard game-based security notions don't necessarily guarantee composability. In fact, many secure protocols might not be secure if several instances are run concurrently.

- QKD and 20(ish) years between first security proofs and composable ones.

- Several frameworks for composability and plenty of work within them, but none have convinced the whole community.

Real-world ideal-world paradigm

AKA simulation paradigm.

Usual definition: a real protocol P securely realizes the ideal functionality F from the resource R if for any attack A on $P \circ R$ there is a simulator S on F such that $(A, P) \circ R$ is indistuingishable from $S \circ F$ by any (efficient) environment.

"Any bad thing that could happen during the protocol could also happen in the ideal world."

Real-world ideal-world paradigm

AKA simulation paradigm.

Usual definition: a real protocol P securely realizes the ideal functionality F from the resource R if for any attack A on $P \circ R$ there is a simulator S on F such that $(A, P) \circ R$ is indistuingishable from $S \circ F$ by any (efficient) environment.

"Any bad thing that could happen during the protocol could also happen in the ideal world."

Usual ways of making this precise:

- Fixing a concrete low-level formalism for interactive computation (e.g. UC-security)
- Abstract cryptography and constructive cryptography — close to our work in spirit but technically different

N+1th approach

In our work we formalize the simulation paradigm over an arbitrary category (and a model of attacks).

N+1th approach

In our work we formalize the simulation paradigm over an arbitrary category (and a model of attacks). The main result is that protocols secure against a fixed attack model can be composed sequentially and in parallel.

N+1th approach

In our work we formalize the simulation paradigm over an arbitrary category (and a model of attacks). The main result is that protocols secure against a fixed attack model can be composed sequentially and in parallel. Some benefits:

N+1th approach

In our work we formalize the simulation paradigm over an arbitrary category (and a model of attacks). The main result is that protocols secure against a fixed attack model can be composed sequentially and in parallel. Some benefits:

- simulation-based security definitions are inherently composable, whether the model of computation is synchronous or not, classical or quantum etc.

N+1th approach

In our work we formalize the simulation paradigm over an arbitrary category (and a model of attacks). The main result is that protocols secure against a fixed attack model can be composed sequentially and in parallel. Some benefits:

- simulation-based security definitions are inherently composable, whether the model of computation is synchronous or not, classical or quantum etc.
- abstract attack models pave way for other kinds of attackers than malicious ones

N+1th approach

In our work we formalize the simulation paradigm over an arbitrary category (and a model of attacks). The main result is that protocols secure against a fixed attack model can be composed sequentially and in parallel. Some benefits:

- simulation-based security definitions are inherently composable, whether the model of computation is synchronous or not, classical or quantum etc.
- abstract attack models pave way for other kinds of attackers than malicious ones
- different notions of security (computational, finite-key regimen etc) fit in

N+1th approach

In our work we formalize the simulation paradigm over an arbitrary category (and a model of attacks). The main result is that protocols secure against a fixed attack model can be composed sequentially and in parallel. Some benefits:

- simulation-based security definitions are inherently composable, whether the model of computation is synchronous or not, classical or quantum etc.
- abstract attack models pave way for other kinds of attackers than malicious ones
- different notions of security (computational, finite-key regimen etc) fit in
- CT and the tools and connections it brings: in particular, string diagrams

Cryptography as a resource theory

The key idea is to view cryptography as a resource theory: the resources are various functionalities (e.g. keys, channels etc) and transformations are given by protocols that build the target resource *securely* from the starting resources.

Cryptography as a resource theory

The key idea is to view cryptography as a resource theory: the resources are various functionalities (e.g. keys, channels etc) and transformations are given by protocols that build the target resource *securely* from the starting resources.

We will discuss an extended example - the OTP. Our approach is inspired by

'*Constructive Cryptography – A New Paradigm for Security Definitions and Proofs*'
Maurer, U., TOSCA 2011.

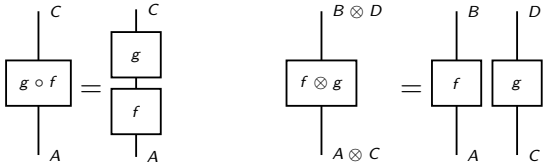
'*Bicategorical Semantics for Nondeterministic Computation*'
Stay, M. & Vicary, J., MFPS 2013.

In this viewpoint, OTP is a protocol: $key \otimes insecure\ channel \to secure\ channel$

Recap on pictures

Let **C** be a symmetric monoidal category — concretely, you can think of (finite) sets and stochastic maps. We will depict a morphism as 

, and composition and monoidal product as



Special morphisms get nicer pictures: identities and symmetries are



Resource theories

Roughly: An SMC where you mostly care whether a hom-set is empty or not.
Examples:

- Can these noisy channels be used to simulate a (almost) noiseless channel?
- Is there a LOCC-protocol that transforms this quantum state to that one?
- Any preordered commutative monoid.

Resource theories

Roughly: An SMC where you mostly care whether a hom-set is empty or not.
Examples:

- Can these noisy channels be used to simulate a (almost) noiseless channel?
- Is there a LOCC-protocol that transforms this quantum state to that one?
- Any preordered commutative monoid.

Many resource theories arise by taking the Grothendieck construction of $\mathbf{D} \xrightarrow{F} \mathbf{C} \xrightarrow{R} \mathbf{Set}$ where F interprets "free operations" in $\mathbf{C}$ and R gives for each $A \in \mathbf{C}$ the set $R(A)$ of resources of type $\mathbf{C}$.

Resource theories

Roughly: An SMC where you mostly care whether a hom-set is empty or not.
Examples:

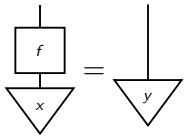
- Can these noisy channels be used to simulate a (almost) noiseless channel?
- Is there a LOCC-protocol that transforms this quantum state to that one?
- Any preordered commutative monoid.

Many resource theories arise by taking the Grothendieck construction of $\mathbf{D} \xrightarrow{F} \mathbf{C} \xrightarrow{R} \mathbf{Set}$ where F interprets "free operations" in $\mathbf{C}$ and R gives for each $A \in \mathbf{C}$ the set $R(A)$ of resources of type $\mathbf{C}$.
Whenever RF is lax symmetric monoidal, $\int RF$ is a symmetric monoidal category, see '*Monoidal Grothendieck construction*'

Moeller & Vasilakopoulou, TAC 2020.

Example resource theories

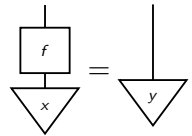
Resource theory of states: apply $\int$ to $\mathbf{C}_F \hookrightarrow \mathbf{C} \xrightarrow{\hom(I,-)} \mathbf{Set}$.
Objects are states of $\mathbf{C}$, and maps $x \to y$ are maps f in $\mathbf{C}_F$ such that



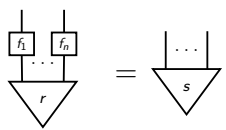
.

Example resource theories

Resource theory of states: apply $\int$ to $\mathbf{C}_F \hookrightarrow \mathbf{C} \xrightarrow{\hom(I,-)} \mathbf{Set}$.
Objects are states of $\mathbf{C}$, and maps $x \to y$ are maps f in $\mathbf{C}_F$ such that



.
n-partite version: apply $\int$ to $\mathbf{C}_F^n \xrightarrow{\otimes} \mathbf{C} \to \mathbf{Set}$. Objects are of the form $((A_i)_{i=1}^n, r\colon I \to \bigotimes A_i)$. A map $(((A_i)_{i=1}^n, r) \to (((B_i)_{i=1}^n, s)$ is then a tuple $(f_i)_{i=1}^n$ that transforms r to s:



We think of this as a resource theory with n-parties who try to agree on actions $f_1, \dots f_n$ to transform some resource to another one.

Security in the abstract

Such a protocol is not necessarily secure—what if some subset of the parties does something else instead?

A general resource theory lets us only talk about correctness of transformations.

Security in the abstract

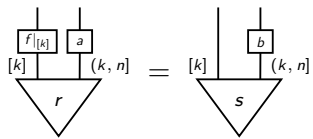
Such a protocol is not necessarily secure—what if some subset of the parties does something else instead?

A general resource theory lets us only talk about correctness of transformations. To add in security:

- need an attack model $\mathcal{A}$ that gives for each protocol f a collection $\mathcal{A}(f)$ of attacks on it
- security against $\mathcal{A}$: for each attack on the protocol there is an attack on the target with similar end-results

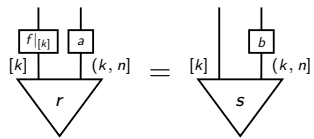
Dishonest parties

Assume the first k parties are honest and the last $n-k$ parties are dishonest. Then $(f_1, \ldots f_n)$ is secure if for any a there is a b such that

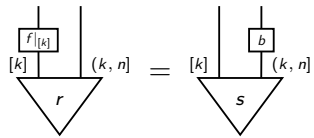


Dishonest parties

Assume the first k parties are honest and the last $n-k$ parties are dishonest. Then $(f_1, \dots f_n)$ is secure if for any a there is a b such that



It suffices to check this for the initial attack $\bigotimes_{k+1}^{n} \mathrm{id}$:



Composability

Theorem

Protocols that are secure against an attack model $\mathcal{A}$ are closed under composition (both $\circ$ and $\otimes$).

Proof.

$\circ$ and $\otimes$ are inherited form the ambient category—one just needs to check that they work.

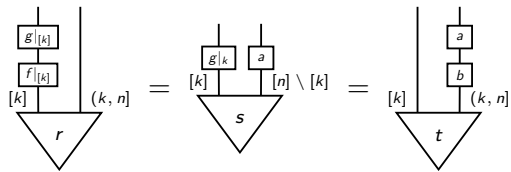
Composability

Theorem

Protocols that are secure against an attack model $\mathcal{A}$ are closed under composition (both $\circ$ and $\otimes$).

Proof.

$\circ$ and $\otimes$ are inherited form the ambient category—one just needs to check that they work. Here's the key steps for $\circ$ and $\otimes$ in the n-partite case with the first k parties honest



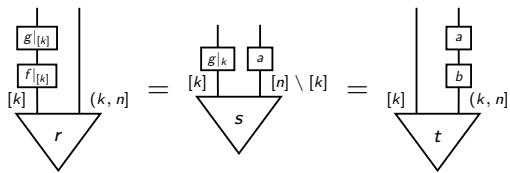
Composability

Theorem

Protocols that are secure against an attack model $\mathcal{A}$ are closed under composition (both $\circ$ and $\otimes$).

Proof.

$\circ$ and $\otimes$ are inherited form the ambient category—one just needs to check that they work. Here's the key steps for $\circ$ and $\otimes$ in the n-partite case with the first k parties honest



□

Security against multiple attack models

Corollary

Protocols secure against $\mathcal{A}_1, \ldots \mathcal{A}_k$ form a symmetric monoidal category

Security against multiple attack models

Corollary

Protocols secure against $\mathcal{A}_1, \ldots \mathcal{A}_k$ form a symmetric monoidal category

Proof.

Symmetric monoidal subcategories are closed under intersection □

Security against multiple attack models

Corollary

Protocols secure against $\mathcal{A}_1, \ldots \mathcal{A}_k$ form a symmetric monoidal category

Proof.

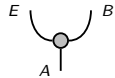
Symmetric monoidal subcategories are closed under intersection □

Example

Fix a family of subsets of n parties: protocols secure against each of these subsets behaving maliciously form an SMC. For instance, in MPC one often studies protocols secure against at most $n/2$ or $n/3$ malicious participants.

OTP: starting resources

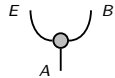
Channel from Alice to Bob that leaks everything to Eve:



(Note: if instead the message goes via Eve (who may tamper with it), the analysis is different)

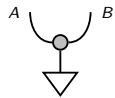
OTP: starting resources

Channel from Alice to Bob that leaks everything to Eve:



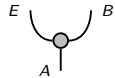
(Note: if instead the message goes via Eve (who may tamper with it), the analysis is different)

Shared random key:



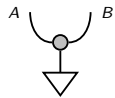
OTP: starting resources

Channel from Alice to Bob that leaks everything to Eve:



(Note: if instead the message goes via Eve (who may tamper with it), the analysis is different)

Shared random key:

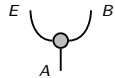


Target resource: a channel



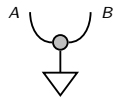
OTP: starting resources

Channel from Alice to Bob that leaks everything to Eve:



(Note: if instead the message goes via Eve (who may tamper with it), the analysis is different)

Shared random key:



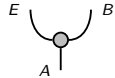
Target resource: a channel



Free building blocks: local (efficient) computation

Insecure protocol

One way of transforming

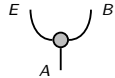


to



Insecure protocol

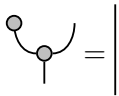
One way of transforming



to



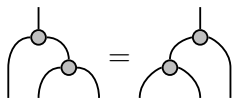
is by having Eve delete everything she receives, as

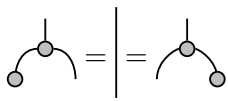


But this is not secure against Eve! No guarantees if Eve disobeys the protocol.

Local ingredients for OTP

A group structure on the message space: a multiplication  with unit  satisfying the following equations.





Local ingredients for OTP

A group structure on the message space: a multiplication  with unit  satisfying the following equations.

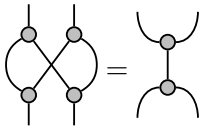


Note that copying and deleting satisfy similar equations



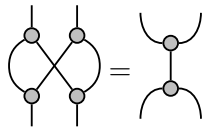
Rest of the group structure

In addition, multiplication and copying interact:

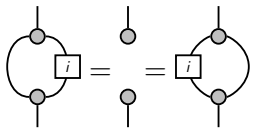


Rest of the group structure

In addition, multiplication and copying interact:

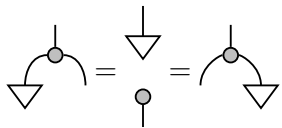


and the map  giving inverses satisfies



Uniform randomness

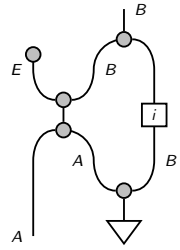
The key being uniformly random is captured by



"Adding uniform noise to a channel gives uniform noise"

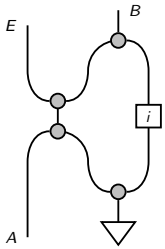
For the experts: a Hopf algebra with an integral in a symmetric monoidal category.

The protocol

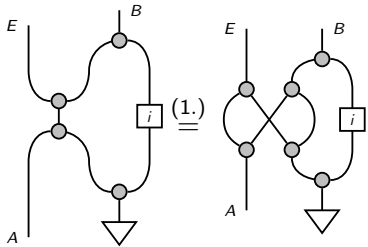


Alice adds the key to her message, broadcasts it to Eve and Bob. Eve deletes her part and Bob adds the inverse of the key to recover the message.

Security of OTP

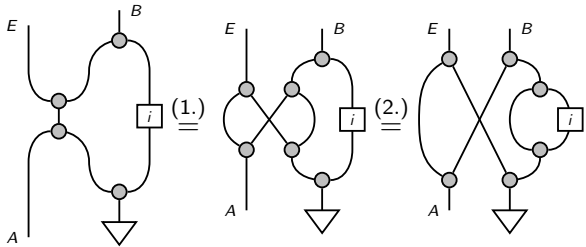


Security of OTP



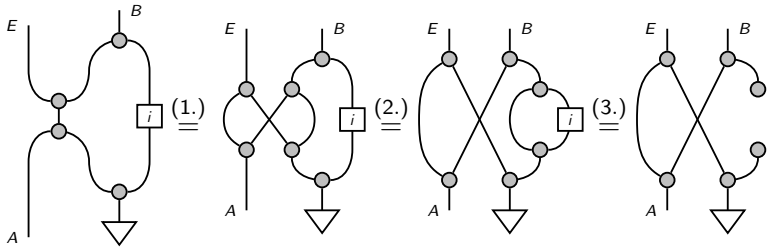
1. Bialgebra.

Security of OTP



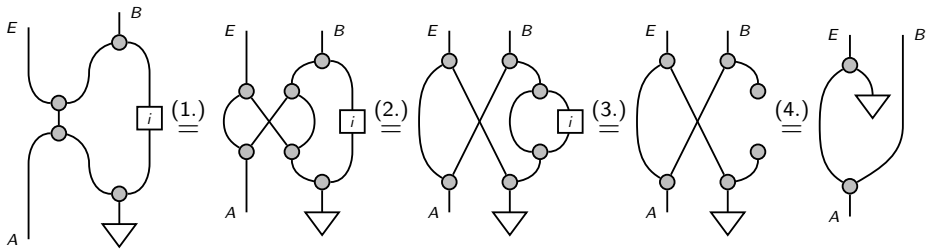
1. Bialgebra. 2. Associativity.

Security of OTP



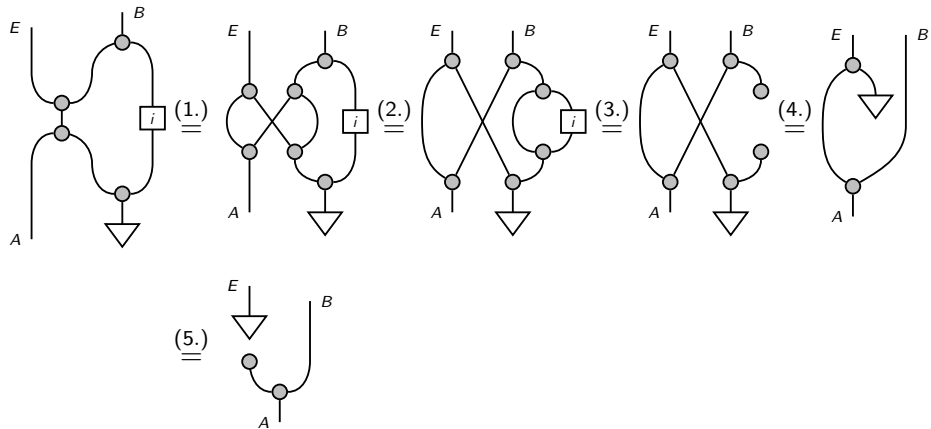
1. Bialgebra. 2. Associativity. 3. Antipode

Security of OTP



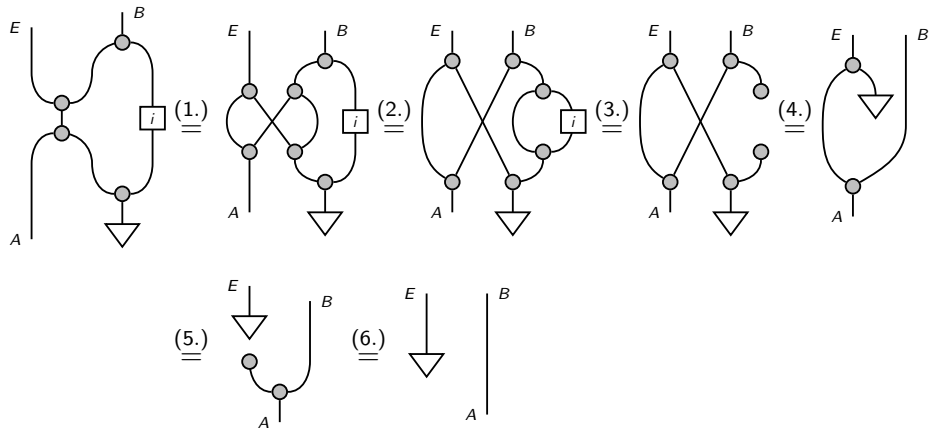
1. Bialgebra. 2. Associativity. 3. Antipode 4. Units

Security of OTP



1. Bialgebra. 2. Associativity. 3. Antipode 4. Units 5. Random noise

Security of OTP



1. Bialgebra. 2. Associativity. 3. Antipode 4. Units 5. Random noise 6. Units.

More on OTP

In other words, anything Eve might learn from the ciphertext she could already compute without it, so this protocol is indeed a secure transformation against Eve.

More on OTP

In other words, anything Eve might learn from the ciphertext she could already compute without it, so this protocol is indeed a secure transformation against Eve.

Reusing keys is not a secure map $key \to key \otimes key$.

More on OTP

In other words, anything Eve might learn from the ciphertext she could already compute without it, so this protocol is indeed a secure transformation against Eve.

Reusing keys is not a secure map $key \to key \otimes key$. However, a computationally secure PRNG will give a computationally secure way of constructing a long shared key from a short one. Composing these two results in *the stream cipher*, which is secure automatically as a composite of secure protocols inside our framework.

More on OTP

In other words, anything Eve might learn from the ciphertext she could already compute without it, so this protocol is indeed a secure transformation against Eve.

Reusing keys is not a secure map $key \to key \otimes key$. However, a computationally secure PRNG will give a computationally secure way of constructing a long shared key from a short one. Composing these two results in *the stream cipher*, which is secure automatically as a composite of secure protocols inside our framework.

(Reasoning about computational security amounts to replacing equations with $\approx$ or working with a (pseudo)metric).

Summary

We have a categorical framework where

- composability is guaranteed (also for computational security)

Summary

We have a categorical framework where

- composability is guaranteed (also for computational security)

- attack models are general enough to cover various kinds of adversarial behavior (e.g. colluding vs independent attackers)

Summary

We have a categorical framework where

- composability is guaranteed (also for computational security)
- attack models are general enough to cover various kinds of adversarial behavior (e.g. colluding vs independent attackers)
- string diagrams can be used to make existing (or new) pictures into rigorous proofs

Questions...

?